Cameron Lee

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Education

Brown University — Master of Science, Computer Science **May 2028 | GPA: 4.0**

San Diego State University — Bachelor of Science, Computer Science **May 2026 | GPA: 3.7**

Skills

Programming: C, C++, C#, Java, Python, Assembly (MIPS, RISC-V), HTML/CSS/JavaScript, SQL, Bash

Platforms/Tools: Linux, Windows, macOS, Git, Unity, ROS 2, Vim, VSCode, Visual Studio, CLion, IntelliJ

Experience

Graduate Research Assistant | Brown University Humans To Robots Lab **October 2026 - Present**

- Researching methods for real-time exocentric VR teleoperation of multiple robots for the "GHOST" project.
- Conducting trials for simultaneous control of Boston Dynamics Spot Robot.
- Using Meta Quest 2 headset with Unity for point cloud visualization and VR control.

Undergraduate Research Assistant | CARE AI Lab **March 2026 - August 2026**

- Assembled and configured PiCar-X robots using Raspberry Pi to create a repeatable testbed for small-scale robotics experiments.
- Investigating automatic PID tuning methods using Python-based control scripts and sensor feedback to increase stability, responsiveness, and navigation performance.
- Supporting experiments on wheeled robots and the Unitree Go1 quadruped, focusing on adaptive control and autonomous behavior in physical environments.

Instructional Assistant & Course Developer | SDSU CS Department **August 2025 - June 2026**

- Designed, revised, and maintained programming assignments in collaboration with faculty for the first three undergraduate programming courses, primarily using C, C++, Java, and Bash to modernize coursework for SDSU's CS curriculum.
- Modernized assignment delivery, automated grading, and feedback workflows for legacy coursework through GitHub/Git, scripting, autograders, unit tests, and LMS integration to reduce manual overhead and improve course scalability.
- Collected student performance data and provided office hours, tutoring, and debugging support to inform course revisions.

Undergraduate Research Assistant | Computer System Intelligence Lab **June 2025 - August 2025**

- Conducted research on methods for quantifying uncertainty in Visual-Inertial Simultaneous Localization and Mapping (VISLAM) systems.
- Analyzed how hardware differences and computational environments impact uncertainty estimates and localization accuracy.

Manager | Abbey's Real Texas BBQ **March 2024 - January 2026**

- Promoted from cashier/server to manager, taking responsibility for supervising staff, coordinating daily operations, and maintaining service quality in a fast-paced environment.
- Designed and implemented a standardized inventory system with checklists and par levels, reducing ordering time by ~50% and virtually eliminating midweek stockouts.
- Streamlined scheduling and daily procedures to increase team efficiency and support more consistent day-to-day execution.

Projects

"Foldaball": Computational Design of Transformable Spheres | **September 2026 - Present**

- Researching methods to automatically optimize arbitrary input geometry into a single-latch spring-loaded sphere that can unfold into a semantically similar shape.
- Uses C++ with optimization algorithms such as Adam and Stochastic Rewrite Descent (SRD) to optimize shape geometry represented with Superfrustra.
- Provides a full end-to-end design system to create a fabricatable design using 3D printing.

F1TENTH Autonomous Racecar | Senior Capstone, Real-Time Embedded Control Lab **August 2025 - July 2026**

- Developed and tuned LiDAR-based reactive planning and control algorithms for a 1/10-scale autonomous racecar using ROS 2, Python, and C++, focusing on disparity-extension obstacle avoidance and Ackermann steering.
- Refined racecar behavior in open straights and narrow environments by iterating on speed-governing PD control, steering smoothing, and gap-selection behavior to reduce oscillation and support stable high-speed navigation.
- Integrated and validated the system in simulated and physical hardware environments with ROS 2 nodes, sensor visualization, and collaborative git workflows; work is continuing beyond the capstone toward a workshop paper.

Linux Kernel Real-Time Scheduling Extensions | Theory of Real Time Systems **September 2025 - December 2025**

- Modified the Linux 4.9 kernel on a Raspberry Pi 3 in C, extending scheduler, timer, and task-management code to support real-time task reservations, periodic execution, and end-to-end latency monitoring.

- Implemented custom kernel system calls and integrated reservation metadata into *task_struct* to support Rate Monotonic and partitioned Earliest Deadline First scheduling models.
- Added per-task execution-time accounting across context switches and automated the cross-compilation, deployment, and testing workflow with Make and shell scripting to accelerate kernel debugging and iteration.

Coda | Personal Project **May 2026 - Present**

- Developed a command-line application in C backed by SQLite to manage and search a personal music collection.
- Implemented database initialization, record creation, editing, listing, and deletion through a flag-based CLI for album metadata.
- Structured the project as a working, documented codebase with support for multiple database files and configurable storage paths.